

Problem 1. Denote by λ the Lebesgue measure on $\mathbb{R}$.

(a) Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous and compactly supported. Show that

$$\lim_{t \rightarrow 0} \int_{\mathbb{R}} |f(x+t) - f(x)| \, d\lambda = 0.$$

(b) Suppose $f \in \mathcal{L}^1(\mathbb{R})$. Show that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f(x) \cos(nx) \, d\lambda = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_{\mathbb{R}} f(x) \sin(nx) \, d\lambda = 0,$$

explaining along the way why the above integrals are well-defined.

Solution. (a) Since the support of f is compact, f is uniformly continuous. Let $U = (a, b)$, with $a, b \in \mathbb{R}$, $a < b$, be an open interval containing the (compact, hence bounded) support of f and let $\varepsilon > 0$. Uniform continuity means there exists $\delta > 0$ such that $|f(x+t) - f(x)| < \varepsilon/|U|$ for all $x, t \in \mathbb{R}$ such that $t < \delta$. Also, $\inf \text{supp } f > a$, so let δ' be the difference $\inf \text{supp } f - a$. If $t < \min\{\delta, \delta'\}$, then

$$\int_{\mathbb{R}} |f(x+t) - f(x)| \, d\lambda = \int_U |f(x+t) - f(x)| \, d\lambda + \int_{\mathbb{R} \setminus U} \underbrace{|f(x+t) - f(x)|}_{0-0=0} \, d\lambda < \int_U \frac{\varepsilon}{|U|} \, d\lambda = \varepsilon,$$

which implies that the limit as t approaches 0 of the left-hand integral is zero.

(b) First, if $b: \mathbb{R} \rightarrow \mathbb{R}$ is a Lebesgue measurable function with $|b(x)| \leq 1$ for all $x \in \mathbb{R}$, then $|fb| \leq |f|$, so

$$\int_{\mathbb{R}} |fb| \, d\lambda \leq \int_{\mathbb{R}} |f| \, d\lambda = \|f\|_1 < \infty, \tag{†}$$

meaning $fb \in \mathcal{L}^1(\mathbb{R})$. Taking $b(x) = \cos(nx), \sin(nx)$ shows that $f(x) \cos(nx)$ and $f(x) \sin(nx)$ are in $\mathcal{L}^1(\mathbb{R})$.

Given $f \in \mathcal{L}^1(\mathbb{R})$, let $(*)$ be the assertion that $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f(x) \cos(nx) \, d\lambda = 0$. We first show that $(*)$ holds if $f = \mathbb{1}_I$ for some interval I with endpoints $a, b \in \mathbb{R}$ where $a < b$. We can compute

$$\left| \int_{\mathbb{R}} \mathbb{1}_I \cos(nx) \, d\lambda \right| = \left| \int_a^b \cos(nx) \, d\lambda \right| = \frac{1}{n} |\sin(nb) - \sin(na)| \leq \frac{2}{n}$$

so, in the limit, the left-hand side is zero, which proves $(*)$. In turn, we can easily see that $(*)$ also holds for “step functions” of the form $\sum_{j=1}^m a_j \mathbb{1}_{I_j}$ where $I_1, \dots, I_m \subseteq \mathbb{R}$ are intervals and $a_1, \dots, a_m \in \mathbb{R}$.

Now, we prove $(*)$ for compactly supported continuous functions f . Let $\varepsilon > 0$; since f is Riemann integrable on an interval I containing $\text{supp } f$ there exists a step function $f_\varepsilon: \mathbb{R} \rightarrow \mathbb{R}$ for which $f_\varepsilon \leq f$ and

$$\|f - f_\varepsilon\|_1 = \left| \int_I f \, d\lambda - \int_I f_\varepsilon \, d\lambda \right| < \varepsilon$$

(so f_ε corresponds to a partition $\mathcal{P}$ of I into intervals such that $L(f, \mathcal{P})$ is within ε of $\int_I f \, d\lambda$). By $(†)$ and the triangle inequality for $\|\bullet\|_1$,

$$\|f \cos(nx)\|_1 \leq \|(f - f_\varepsilon) \cos(nx)\|_1 + \|f_\varepsilon \cos(nx)\|_1 \leq \|f - f_\varepsilon\|_1 + \|f_\varepsilon \cos(nx)\|_1 < \varepsilon + \|f_\varepsilon \cos(nx)\|_1. \quad (**)$$

Taking the limit, we have via (*) for step functions that for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \|f \cos(nx)\|_1 \leq \varepsilon + \lim_{n \rightarrow \infty} \|f_\varepsilon\|_1 \stackrel{(*)}{=} \varepsilon.$$

This shows that the limit on the left is zero, proving (*) for compactly supported continuous functions.

Finally, let $f \in \mathcal{L}^1(\mathbb{R})$ be arbitrary. Let $\varepsilon > 0$; by Axler's 3.48, there exists a compactly supported continuous function $g_\varepsilon: \mathbb{R} \rightarrow \mathbb{R}$ such that $\|f - g_\varepsilon\|_1 < \varepsilon$. Then, the inequality (**) holds here with g_ε in place of f_ε , which as above implies that

$$\lim_{n \rightarrow \infty} \|f \cos(nx)\|_1 \leq \varepsilon + \lim_{n \rightarrow \infty} \|g_\varepsilon\|_1 \stackrel{(*)}{=} \varepsilon \implies \lim_{n \rightarrow \infty} \|f \cos(nx)\|_1 = 0,$$

this time using (*) for compactly supported continuous functions to extract (*) in the generic case.

The last thing to do is to prove the corresponding result with sin. Since $\sin(x) = \cos(\frac{\pi}{2} - x) = \cos(x - \frac{\pi}{2})$ and Lebesgue measure is translation invariant,

$$\int_{\mathbb{R}} f(x) \sin(nx) \, d\lambda = \int_{\mathbb{R}} f\left(x - \frac{\pi}{2n}\right) \sin\left(n\left(x - \frac{\pi}{2n}\right)\right) \, d\lambda = \int_{\mathbb{R}} f\left(x - \frac{\pi}{2n}\right) \cos(nx) \, d\lambda.$$

If $f_n(x) := f(x - \frac{\pi}{2n})$ then, similarly to the first part of the bound in (**) (with f_n in place of f_ε),

$$\begin{aligned} \lim_{n \rightarrow \infty} \|f \sin(nx)\| &= \lim_{n \rightarrow \infty} \|f_n \cos(nx)\|_1 \\ &\leq \lim_{n \rightarrow \infty} \|f_n \cos(nx) - f \cos(nx)\|_1 + \underbrace{\lim_{n \rightarrow \infty} \|f \cos(nx)\|_1}_0 \leq \lim_{n \rightarrow \infty} \|f - f_n\| \stackrel{(a)}{=} 0. \quad \square \end{aligned}$$

Problem 2 (Axler 4A #1,2). (a) Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $f: X \rightarrow \mathbb{R}$ is an $\mathcal{S}$ -measurable function. Prove that

$$\mu(\{x \in X : |f(x)| \geq c\}) \leq \frac{1}{c^p} \int_X |f|^p \, d\mu$$

for all positive numbers c and p .

(b) Suppose $(X, \mathcal{S}, \mu)$ is a measure space with $\mu(X) = 1$ and $f \in \mathcal{L}^1(\mu)$. Prove that

$$\mu\left(\left\{x \in X : \left|f(x) - \int_X f \, d\mu\right| \geq c\right\}\right) \leq \frac{1}{c^2} \left(\int_X f^2 \, d\mu - \left(\int_X f \, d\mu\right)^2\right).$$

Solution. (a) Since c and p are positive, $|f(x)| \geq c$ if and only if $|f(x)|^p \geq c^p$; thus,

$$c^p \mu(\{x \in X : |f(x)| \geq c\}) = c^p \mu(\{x \in X : |f(x)|^p \geq c^p\}) = \int_X c^p \mathbb{1}_{\{x \in X : |f(x)|^p \geq c^p\}} \, d\mu \leq \int_X |f|^p \, d\mu.$$

(b) Let $M = \int_X f \, d\mu$. Using (a), we can compute

$$\begin{aligned} \mu(\{x \in X : |f(x) - M| \geq c\}) &\leq \frac{1}{c^2} \int_X (f - M)^2 \, d\mu = \frac{1}{c^2} \int_X (f^2 - 2Mf + M^2) \, d\mu \\ &= \frac{1}{c^2} \left(\int_X f^2 \, d\mu - 2M \int_X f \, d\mu + \int_X M^2 \, d\mu\right) \\ &= \frac{1}{c^2} \left(\int_X f^2 \, d\mu - 2M^2 + M^2 \mu(X)\right) = \frac{1}{c^2} \left(\int_X f^2 \, d\mu - M^2\right). \quad \square \end{aligned}$$

Problem 3 (Axler 4A #12,13). (a) Show that $|\{b \in \mathbb{R} : f^*(b) = \infty\}| = 0$ for every $f \in \mathcal{L}^1(\mathbb{R})$.

(b) Show that there exists $f \in \mathcal{L}^1(\mathbb{R})$ such that $f^*(b) = \infty$ for every $b \in \mathbb{Q}$.

Solution. (a) Since $\{b \in \mathbb{R} : f^*(b) = \infty\} = \bigcap_{n=1}^{\infty} \{b \in \mathbb{R} : f^*(b) > n\}$, Axler's 4.8 implies that for all $n \in \mathbb{N}$,

$$|\{b \in \mathbb{R} : f^*(b) = \infty\}| \leq |\{b \in \mathbb{R} : f^*(b) > n\}| \leq \frac{3}{n} \|f\|_1,$$

which means $|\{b \in \mathbb{R} : f^*(b) = \infty\}| = 0$.

(b) Let $\{q_1, q_2, \dots\} = \mathbb{Q}$ and, for $n, k \in \mathbb{N}$, let $I_{n,k} = (q_n - 2^{-k}, q_n + 2^{-k})$. Then, define $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$$f_n = 2^{-n} \sum_{k=1}^{\infty} \mathbb{1}_{I_{n,k}}.$$

For any $m \in \mathbb{N}$, since $I_{n,m}$ is centered at q_n ,

$$f_n^*(q_n) \geq \frac{1}{|I_{n,m}|} \int_{I_{n,m}} f_n \, d\lambda \geq \frac{1}{|I_{n,m}|} \int_{I_{n,m}} \left(2^{-n} \sum_{k=1}^m \mathbb{1}_{I_{n,k}} \right) \, d\lambda = \frac{1}{|I_{n,m}|} \cdot 2^{-n} \cdot m \cdot |I_{n,m}| = 2^{-n} m,$$

which implies that $f_n^*(q_n) = \infty$. Now define $f = \sum_{n=1}^{\infty} f_n$. This function is in $\mathcal{L}^1(\mathbb{R})$ because

$$\int_{\mathbb{R}} f \, d\mu = \sum_{n=1}^{\infty} \int_{\mathbb{R}} f_n \, d\mu = \sum_{n=1}^{\infty} \left(2^{-n} \sum_{k=1}^{\infty} |I_{n,k}| \right) = \sum_{n=1}^{\infty} \left(2^{-n} \sum_{k=1}^{\infty} 2^{-k+1} \right) = \sum_{n=1}^{\infty} 2^{-n} \cdot 2 = 2 < \infty.$$

Furthermore, since $0 \leq f_n \leq f$ for all n ,

$$f^*(q_n) = \sup_{t \geq 0} \frac{1}{2t} \int_{[q_n-t, q_n+t]} f \, d\lambda \geq \sup_{t \geq 0} \frac{1}{2t} \int_{[q_n-t, q_n+t]} f_n \, d\lambda = f_n^*(q_n) = \infty,$$

which means f^* is ∞ on $\mathbb{Q}$, so f has the desired properties. □

Problem 4 (Axler 4A #9, 4B#5). (a) Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is Lebesgue measurable. Prove that

$$\{b \in \mathbb{R} : f^*(b) > c\}$$

is an open subset of $\mathbb{R}$ for every $c \in \mathbb{R}$.

(b) Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a Lebesgue measurable function. Prove that $|f(b)| \leq f^*(b)$ for almost every $b \in \mathbb{R}$.

Solution. (a) We show the equivalent statement that $A = \{b \in \mathbb{R} : f^*(b) \leq c\}$ is closed in $\mathbb{R}$. Let $b_1, b_2, \dots \in A$ be a sequence of points converging to $b \in \mathbb{R}$. Let U be an open interval $(b-r, b+r)$ centered at b with radius r and for each $n \in \mathbb{N}$, let U_n be the smallest open interval containing U that is centered at b_n , i.e. $U_n = (b_n - (r + |b - b_n|), b_n + (r + |b - b_n|))$. Since $b_n \in A$, we can bound $\int_{U_n} |f| \, d\lambda \leq c |U_n|$, so

$$\frac{1}{|U|} \int_U |f| \, d\lambda \leq \frac{1}{|U|} \int_{U_n} |f| \, d\lambda \leq \frac{1}{|U|} \cdot c |U_n|$$

for all $n \in \mathbb{N}$. Since b_n converges to b , $|U_n| = |U| + 2(b - b_n)$ converges to $|U|$, so the right-hand side converges to $\frac{1}{|U|} \cdot c |U| = c$, meaning the left-hand side is bounded above by c for every open interval U centered at b . This means $b \in A$, so A contains all of its limit points and must be closed.

(b) If $f^*(b) = \infty$ everywhere then the inequality holds automatically. Otherwise, there exists b such that $f^*(b) < \infty$, meaning $\frac{1}{2t} \int_{(b-t, b+t)} |f| \, d\lambda$ is finite for all $n \in \mathbb{N}$. Thus, if $U_n = (b-n, b+n)$, then $f|_{U_n} \in \mathcal{L}^1(\mathbb{R})$. By Axler's 4.21 on $f|_{U_n}$,

$$|f(b)| = |f|_{U_n}(b) = \lim_{t \rightarrow 0^+} \frac{1}{2t} \int_{(b-t, b+t)} |f|_{U_n}(b) \, d\lambda \leq \sup_{t > 0} \frac{1}{2t} \int_{(b-t, b+t)} |f(b)| \, d\lambda = f^*(b).$$

for almost all $b \in U_n$; let U_n^* be the set of all such b . Then, $\bigcup_{n=1}^{\infty} U_n^*$ contains almost every point of $\mathbb{R}$ because its complement is the intersection of measure-zero sets $\bigcap_{n=1}^{\infty} (\mathbb{R} \setminus U_n^*)$; this finishes the proof. $\square$

Problem 5. Let $A \subseteq \mathbb{R}$ be a Lebesgue measurable subset and denote by λ the Lebesgue measure on $\mathbb{R}$.

- (a) Suppose $|A| < \infty$ and $a \in \mathbb{R}$ is such that the density $\delta_A(a) \in [0, 1]$ of A at a is defined and $\delta_A(a) \in \{0, 1\}$. Define $g_A: \mathbb{R} \rightarrow \mathbb{R}$ by

$$g_A(x) = \int_{(-\infty, x)} \mathbb{1}_A \, d\lambda.$$

Show that $g'_A(a)$ exists and equals $\delta_A(a)$.

- (b) Suppose $\delta_A(a) = 1$ for every $a \in A$ and $\delta_A(a) = 0$ for every $a \in \mathbb{R} \setminus A$. Show that $A = \emptyset$ or $A = \mathbb{R}$.

Solution. (a) If $\delta_A(a) = 0$, then the one-sided limits

$$\lim_{t \rightarrow 0^+} \frac{|A \cap (a-t, a)|}{2t}, \quad \lim_{t \rightarrow 0^+} \frac{|A \cap (a, a+t)|}{2t}$$

must both exist and equal 0 because we can bound

$$0 \leq \liminf_{t \rightarrow 0^+} \frac{|A \cap (a-t, a)|}{2t} \leq \limsup_{t \rightarrow 0^+} \frac{|A \cap (a-t, a)|}{2t} \leq \limsup_{t \rightarrow 0^+} \frac{|A \cap (a-t, a+t)|}{2t} = \delta_A(a) = 0$$

and similarly with $\frac{|A \cap (a, a+t)|}{2t}$. This implies immediately that $g'_A(a)$ exists and equals $\delta_A(a) = 0$.

If $\delta_A(a) = 1$, then for any $\varepsilon > 0$, there exists $\delta > 0$ such that $0 < t < \delta$ implies

$$1 - \varepsilon < \frac{|A \cap (a-t, a+t)|}{2t} = \frac{|A \cap (a-t, a)|}{2t} + \frac{|A \cap (a, a+t)|}{2t}.$$

The two addends on the right are both bounded above by $\frac{1}{2}$, so they must be bounded below by $\frac{1}{2} - \varepsilon$; therefore,

$$\frac{1}{2} - \varepsilon \leq \liminf_{t \rightarrow 0^+} \frac{|A \cap (a-t, a)|}{2t} \leq \liminf_{t \rightarrow 0^+} \frac{|A \cap (a, a+t)|}{2t} \leq \frac{1}{2}.$$

Since this holds for every $\varepsilon > 0$, we get $\lim_{t \rightarrow 0^+} \frac{|A \cap (a-t, a)|}{2t} = \frac{1}{2}$. By the same logic $\lim_{t \rightarrow 0^+} \frac{|A \cap (a, a+t)|}{2t} = \frac{1}{2}$. The existence and equality of these one-sided limit implies that $g'_A(a)$ exists and equals $\delta_A(a)$.

- (b) Let U_a be an open interval centered at $a \in A$ so that $A \cap U_a$ has finite measure, and then define

$$g_a(x) = \int_{(-\infty, x)} \mathbb{1}_{A \cap U_a} \, d\lambda.$$

By (a) and the given information about δ_A , $g'_a(x) = \delta_{A \cap U_a}(x) = \delta_A(x) = \mathbb{1}_A(x)$ for all $x \in U_a$.

Recall the intermediate value theorem for derivatives:

Theorem (IVT for derivatives). Let $I \subseteq \mathbb{R}$ be an open interval and let $f: I \rightarrow \mathbb{R}$ be differentiable on I . If $a, b \in I$ are such that $f'(a) < f'(b)$, then the interval $[f'(a), f'(b)]$ is contained in the image $f'(I)$.

We know that $g' = \mathbb{1}_A|_{U_a}$ only takes on the values 0 and 1, so according to the theorem above, it must be constant on U_a , since otherwise g' would have to take on every value in $[0, 1]$. This means $\mathbb{1}_A$ is constant on each U_a , i.e. it is locally constant, hence constant because $\mathbb{R}$ is connected. Therefore, $\mathbb{1}_A$ is either 0 everywhere, in which case $A = \emptyset$, or $\mathbb{1}_A$ is 1 everywhere, in which case $A = \mathbb{R}$. $\square$

Problem 6. Let $A, B \subseteq \mathbb{R}$ be Lebesgue measurable subsets.

- (a) Suppose $a, b \in \mathbb{R}$ are such that the densities $\delta_A(a), \delta_B(b) \in [0, 1]$ of A at a and B at b , respectively, are defined and $\delta_A(a) + \delta_B(b) > 1$. Show that the subset

$$A + B := \{x + y : x \in A, y \in B\}$$

contains an open interval around $a + b$.

- (b) Show that for almost all $a \in A$ and almost all $b \in B$ the subset $A + B \subseteq \mathbb{R}$ contains an open interval around $a + b$.

Solution. (a) Since Lebesgue measure is translation invariant, we can assume by translating A and B that $a = b = 0$. Then, supposing for the sake of contradiction that $A + B$ does not contain an open interval around $a + b = 0$, let $\varepsilon > 0$ and let $t_\varepsilon \in (-\varepsilon, \varepsilon)$ be a number not contained in $A + B$.

Let $t > 0$. If $x \in A \cap (-t, t)$ then $t_\varepsilon - x \in B^c \cap (-t - t_\varepsilon, t + t_\varepsilon)$ where $B^c = \mathbb{R} \setminus B$, so

$$|B^c \cap (-t - t_\varepsilon, t + t_\varepsilon)| \geq |\{t_\varepsilon - x : x \in A \cap (-t, t)\}| = |A \cap (-t, t)|$$

where the equality comes from translation invariance. This implies that

$$\begin{aligned} |B \cap (-t, t)| &\leq |B \cap (-t - t_\varepsilon, t + t_\varepsilon)| \leq 2(t + t_\varepsilon) - |B^c \cap (-t - t_\varepsilon, t + t_\varepsilon)| \leq 2(t + \varepsilon) - |A \cap (-t, t)| \\ &\iff |A \cap (-t, t)| + |B \cap (-t, t)| < 2t + 2\varepsilon \\ &\iff \frac{|A \cap (-t, t)| + |B \cap (-t, t)|}{2t} \leq 1 + \frac{\varepsilon}{t}. \end{aligned}$$

Since this holds for all $\varepsilon > 0$, this means the upper bound can be strengthened to 1, so

$$\delta_A(0) + \delta_B(0) = \lim_{t \rightarrow 0^+} \frac{|A \cap (-t, t)| + |B \cap (-t, t)|}{2t} \leq 1,$$

which is a contradiction, so $A + B$ contains an open interval around 0.

- (b) By Axler's 4.24, almost every $a \in A$ has $\delta_A(a) = 1$ and almost every $b \in B$ has $\delta_B(b) = 1$. We are then done by (a), since if $\delta_A(a) = 1$ and $\delta_B(b) = 1$ (hence $\delta_A(a) + \delta_B(b) > 1$), $A + B$ contains an open interval around $a + b$. $\square$